Stochastic Processes Semester 1

Tutorial 7 Extension Questions – Solutions

1. (a) The probability could be calculated through following equation:

$$\mathbf{P}(N(18;22) = 0 \mid N(60) = 1500)$$

if $\lambda = 30$. Again, please note that the probability does not depend on λ.

(b) Similarly to (a), this probability could be calculate through following equation:

$$\mathbf{P}(N(40;45) = 10 \mid N(60) = 1500).$$

(c) i. Let I be the interval $(5.40, 5.45)$ and $N_f(I)$, $N_m(I)$ be the number of, respectively, female and male arrivals in this interval, so we want to find $\mathbf{P}(N_f(I) = 4 \mid N(I) \leq 5)$. Then

$$\begin{aligned}
\mathbf{P}(N_f(I) = 4 \mid N(I) \leq 5) &= \frac{\mathbf{P}(N_f(I) = 4, N(I) \leq 5)}{\mathbf{P}(N(I) \leq 5)} \\
&= \frac{\mathbf{P}(N_f(I) = 4, N_m(I) \leq 1)}{\mathbf{P}(N(I) \leq 5)} \\
&= \frac{\mathbf{P}(N_f(I) = 4)\mathbf{P}(N_m(I) \leq 1)}{\mathbf{P}(N(I) \leq 5)} \\
&= \frac{(3\lambda)^4 e^{-(3\lambda)}/4! \left(e^{-(2\lambda)} + (2\lambda)e^{-(2\lambda)}\right)}{\sum_{k=0}^{5} (5\lambda)^k e^{-5\lambda}/k!}.
\end{aligned}$$

Again, the probability does depend on the time interval.

(c) ii. Now the probability given as:

$$\begin{aligned}
\mathbf{P}(N_f(I) = 4 \mid N(I) \geq 5) &= \frac{\mathbf{P}(N_f(I) = 4, N(I) \geq 5)}{\mathbf{P}(N(I) \geq 5)} \\
&= \frac{\mathbf{P}(N_f(I) = 4, N_m(I) \geq 1)}{\mathbf{P}(N(I) \geq 5)} \\
&= \frac{\mathbf{P}(N_f(I) = 4)\mathbf{P}(N_m(I) \geq 1)}{\mathbf{P}(N(I) \geq 5)} \\
&= \frac{(3\lambda)^4 e^{-(3\lambda)}/4! \left(1 - e^{-(2\lambda)}\right)}{(1 - \sum_{k=0}^{4} (5\lambda)^k e^{-5\lambda}/k!)}.
\end{aligned}$$

2. (a) We require the probability that a Poisson random variable with mean 15 (3 per minute, 5 minutes) takes the value 0, which is $e^{-15} \approx 3.06 * 10^{-7}$.

(b) In the case $\alpha = 0.4$,

$$\begin{aligned}
\mathbf{P}(N_F[0,5] > 2 | N_M[0,5] > 4) &= \mathbf{P}(N_F[0,5] > 2) \quad \text{(independence)} \\
&= 1 - \mathbf{P}(N_F[0,5] \leq 2) \\
&= 0.323 \quad \text{since } N_F[0,5] \text{ has a Poisson(2) dist.}
\end{aligned}$$

(c) The time between Joe's successive conversations is the sum of 5 independent Exp(1) random variables, which is a gamma distribution with shape parameter 5 and scale parameter 1. Alternatively, the required probability is the probability that a Poisson random variable with mean 5 takes the value at most 4, which is 0.4405.

(d) The probability will decrease when the customers arriving rate increasing.

3. (a) This is the probability that a binomial random variable with parameters 10 and 2/3 takes the value at least 6, which is 0.787.

   (b) Now the x is

$$x = \frac{2}{3}x^2 + \frac{1}{3}.$$